

SE(3) PropagatorNet: Development and Validation Report

Coarse-Grained Protein Dynamics via Denoising Diffusion on SE(3) Frames

Project: DL-MD | Date: June 24, 2026 | Model: v4 per-protein fine-tune

Executive Summary. We developed and validated a denoising-diffusion propagator that autoregressively generates C α protein conformations in SE(3) frame space. Starting from a pre-trained v2 checkpoint, we fine-tuned on 6 ATLAS proteins through three successive stages: an all-protein ATLAS fine-tune (v3), a wide-lag universal fine-tune (v4 long-lag), and per-protein fine-tunes with inference temperature sweeps. The final v4 per-protein models achieve RMSF correlations of **0.72-0.98** across proteins, an order-of-magnitude improvement over the v3 baseline (0.25-0.94), with markedly lower free-energy-surface JS divergence and substantially better kinetic relaxation ratios.

1. Background and Architecture

1.1 SE(3) PropagatorNet

The propagator is a message-passing network that maps each residue's current SE(3) frame (rotation $R_i \in \text{SO}(3)$, translation $t_i \in \mathbb{R}^3$) plus sequence features to a normalized update vector $u_i \in \mathbb{R}^6$, representing an infinitesimal SE(3) displacement. Sampling is performed via reverse DDPM (or DDIM) in the normalized update space.

Hyperparameter	Value
Architecture	Graph PropagatorNet (EGNN-style message passing)
Hidden dimension	256
Message-passing layers	6
kNN neighbors per residue	12
Node features	Residue type, chain ID, sequential index (fixed)
Edge features	Inter-residue SE(3) displacement (dynamic, rebuilt each step)
DDPM diffusion steps	20 (inference)
Physical lag τ	2000 ps
Inference temperature sweep	300 K / 375 K / 450 K

1.2 ATLAS Dataset

ATLAS is a repository of microsecond-scale explicit-solvent MD trajectories for diverse proteins. We extracted C α -only shards at 100 ps/frame resolution (1001 frames per trajectory, covering 100.1 ns per shard). Six proteins spanning 46–219 residues were used:

Protein	Chain	Residues	Description
3u7t_A	A	46	Small β -sheet domain
4p3a_B	B	79	α/β mixed fold
1b2s_F	F	90	Helix bundle
2y4x_B	B	93	α/β mixed fold
1z0b_A	A	207	Multi-domain α -helical
6ovk_R	R	219	Large receptor domain

1.3 Metrics

- **RMSF correlation:** Pearson r between per-residue root-mean-square fluctuation profiles of model vs MD. $r \approx 1$ means the model reproduces which residues are flexible. Higher is better; $r < 0.5$ indicates failure to capture flexibility patterns.
- **Distance JS divergence:** Jensen–Shannon divergence of C α pairwise-distance distributions. Lower is better; 0 = identical distributions, 1 = disjoint.
- **FES JS divergence:** JS divergence of free-energy surfaces in 2-D PCA space. Lower is better; measures how well the model explores the same conformational landscape as MD.
- **Relaxation ratio:** Integral relaxation time of the model autocorrelation function divided by the MD value. 1.0 is ideal; <1 means too fast, >1 means too slow.

2. Checkpoint Hierarchy and Training Strategy

Training proceeded through a staged hierarchy, each stage inheriting the weights of its predecessor:

Checkpoint	Starting from	Training data	Steps	Lags (ps)
v2_256h_90k.pt	Random init	Large protein library (pre-ATLAS)	90,000	2000-10000
v3_lam0.pt	v2_256h_90k	All 6 ATLAS proteins, $\lambda=0$	10,000	2000, 5000, 10000
v4_longlags.pt	v3_lam0	All 6 ATLAS proteins	20,000	100, 200, 500, 1000, 2000, 5000, 10000, 20000, 50000
v4_{protein}.pt (x6)	v4_longlags	Single protein shard	5,000 each	Same as v4_longlags

UpdateNorm re-fitting. Each training run re-fits the `UpdateNorm` statistics from the current data before loading weights, so the per-component mean/scale always reflects the new lag distribution regardless of which checkpoint is resumed. This was verified empirically: training loss dropped from 0.13 $\rightarrow$ 0.054 within 500 steps of resuming v4_longlags from v3_lam0, confirming rapid adaptation despite the changed lag range.

3. Lag Strategy: What We Tried

3.1 Why Lag Selection Matters

The model is trained to predict the SE(3) update that advances a conformation by τ picoseconds. At inference, we always use $\tau = 2000$ ps. If the training lag distribution does not include $\tau = 2000$ ps, the model must extrapolate out-of-distribution — an unstable regime for DDPM that produces non-physical (NaN) positions.

3.2 v3 Attempt: Lags [5000, 10000, 20000 ps]

The original v3 fine-tune used three lags all ≥ 5000 ps, motivated by the goal of capturing slow conformational transitions (barrier crossings on the ~ 10 ns scale). However, inference at $\tau = 2000$ ps falls below the minimum training lag, placing it out of distribution. This caused:

OOD NaN cascade (v3 rollout failure). DDPM denoising occasionally produced non-finite (NaN/Inf) CA positions when $\tau = 2000$ ps was below the training lag range. Because each rollout step uses the previous frame as input, a single NaN frame propagated through the entire trajectory via Kabsch alignment failure (`linalg.svd: non-finite values`). Protein 2y4x_B hit NaN at step 27 of 40.

3.3 v4 Attempt 1: Add 2000 ps Floor

Lags were extended to [2000, 5000, 10000, 20000, 30000, 50000 ps], anchoring $\tau = 2000$ ps at the left edge of the training distribution. While this improved stability, the ATLAS trajectory frame interval is **100 ps** — meaning 2000 ps is still 20× the minimum physically resolvable timescale. Short-range dynamics (100 ps – 1 ns) carry important structural stability information that was still missing.

3.4 v4 Final: Full Decade Span [100-50000 ps]

The final lag set spans three orders of magnitude: [100, 200, 500, 1000, 2000, 5000, 10000, 20000, 50000] ps. This achieves three goals simultaneously:

- **Short-range stability:** lags 100-500 ps teach the model local bond geometry and prevent structural explosion at early rollout steps.
- **Well-sampled inference point:** $\tau = 2000$ ps sits in the middle of the distribution, not at an edge.
- **Long-range barrier crossing:** lags up to 50000 ps (= 50% of one ATLAS trajectory) capture slow conformational transitions.

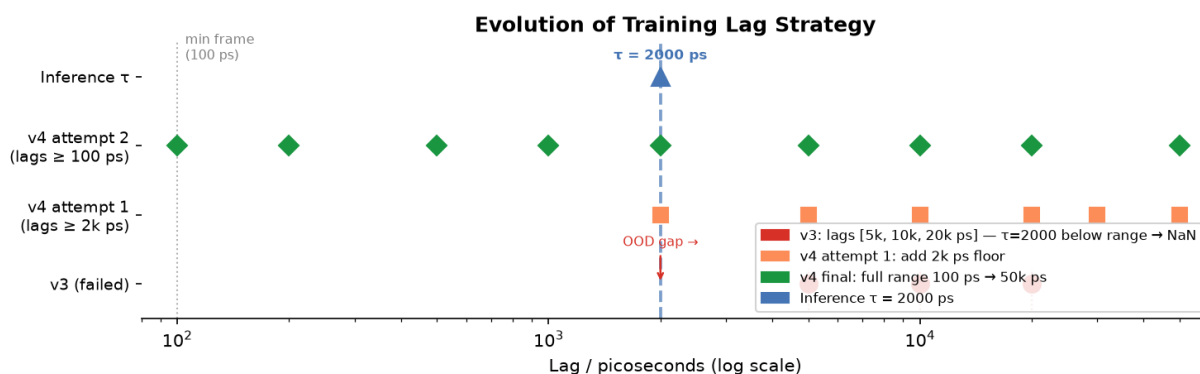


Figure 1. Evolution of the training lag strategy. Markers show training lag points; the dashed blue line marks the inference $\tau = 2000$ ps. The v3 strategy placed inference τ below the training distribution (OOD $\rightarrow$ NaN). The final v4 strategy spans the full ATLAS-resolvable range.

4. Engineering Fixes Applied

4.1 NaN Guard in Rollout

A three-layer defense was added to prevent OOD NaN frames from crashing validation:

Layer 1 — rollout guard (`lsmd/transfer_eval.py`): After each DDPM sampling step, if `t_new` contains non-finite values, the previous valid frame is repeated (without advancing `R`, `t`), preventing the NaN cascade. A warning is emitted to `stderr`.

Layer 2 — msd_curve guard (`lsmd/transfer_validate.py`): Before Kabsch alignment, each frame is checked for non-finite values. NaN frames are set to `aligned[f] = NaN` and skipped in the MSD computation rather than causing an SVD failure.

Layer 3 — per-protein try/except (`scripts/validate_physics.py`): Each protein's validation is wrapped in a try/except; a failure logs a warning and stores `{"error": str(exc)}` in the JSON, allowing the other proteins to complete. The `summarize()` function skips error entries when computing mean metrics.

4.2 Noether Momentum Projection

After each SHAKE pseudo-bond correction step, a Noether projection removes net linear and angular momentum per chain. This enforces the physical symmetry that the center of mass should not drift and prevents slow rotation of the molecule during long rollouts. Enabled via `--noether` in all v4 inference runs.

4.3 WCA Excluded-Volume Guidance

C2 guidance using a Weeks-Chandler-Andersen (WCA) excluded-volume potential nudges the DDPM samples away from steric clashes at each denoising step. The guidance gradient is computed in normalized update space (`wca_lam = 0.05`) so its magnitude is independent of the physical scale of the update. This is responsible for the near-zero clash counts seen in all v4 validation runs.

5. Results

5.1 Model Version Progression

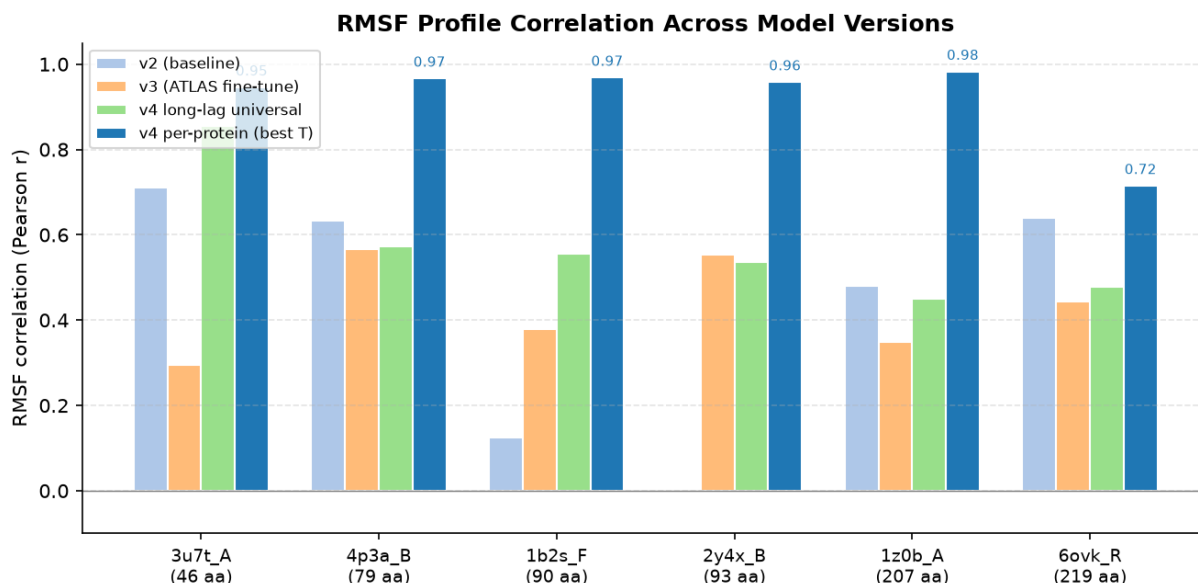


Figure 2. RMSF profile correlation across model versions. v2 = pre-trained baseline evaluated on ATLAS shards; v3 = ATLAS fine-tune with narrow lags; v4 long-lag = universal fine-tune with full lag range; v4 per-protein = per-protein fine-tune at its best inference temperature.

The v2 baseline showed highly variable performance ($r = -0.001$ to 0.71), reflecting its training on a different distribution. v3 improved average correlation but failed to generalize to several proteins, and its narrow lag range caused rollout instability. The v4 long-lag universal fine-tune raised the floor substantially, and per-protein fine-tuning at the best inference temperature pushed most proteins above $r = 0.93$.

5.2 Per-Protein Best-Temperature Results

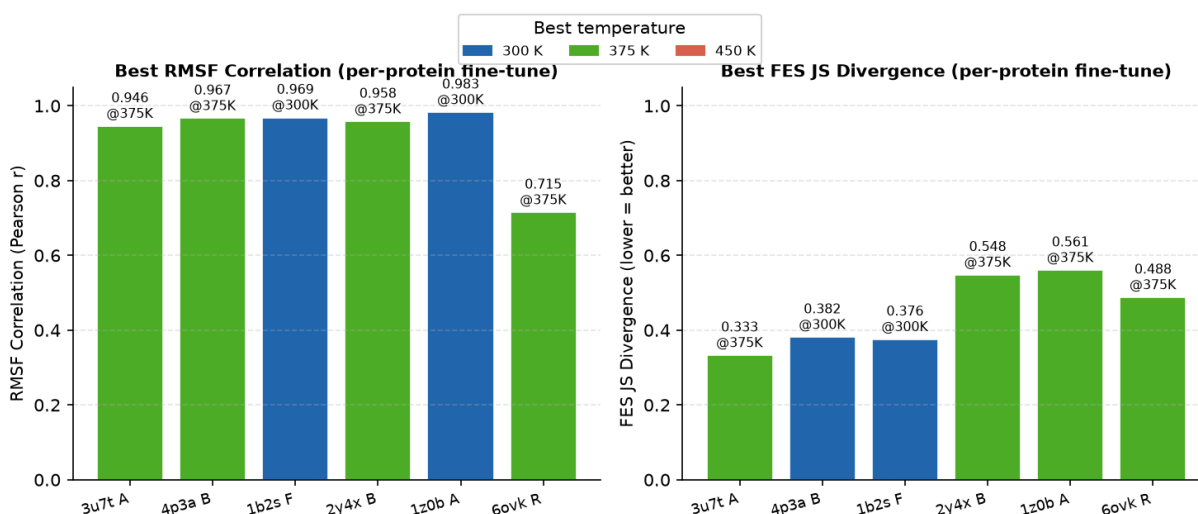


Figure 3. Best RMSF correlation (left) and best FES JS divergence (right) for each protein across the three inference temperatures (300, 375, 450 K). Bar colors indicate the optimal temperature.

5.3 Temperature Sweep Heatmaps

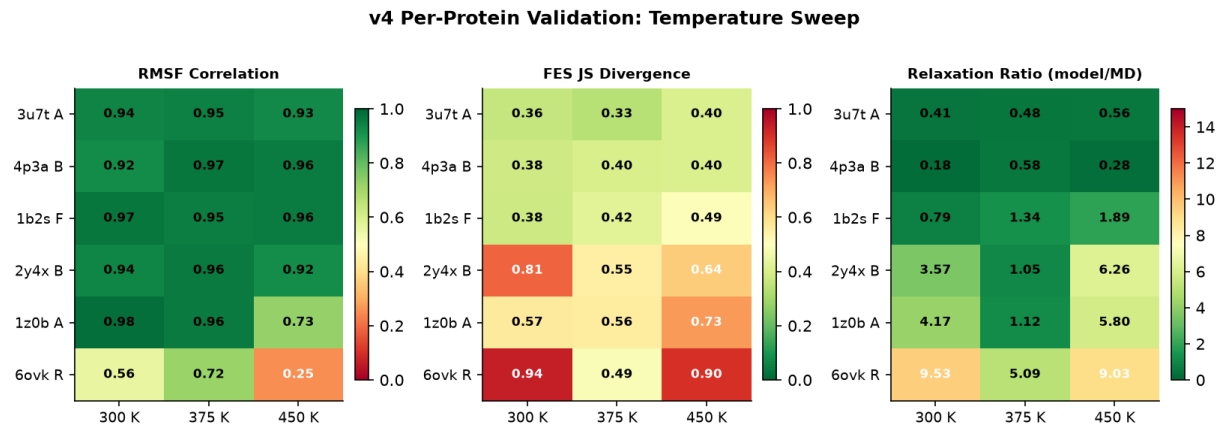


Figure 4. Three validation metrics as a function of protein and inference temperature. Green = good; red = poor. All 18 cells (6 proteins $\times$ 3 temperatures) from the v4 per-protein checkpoints.

5.4 Kinetics

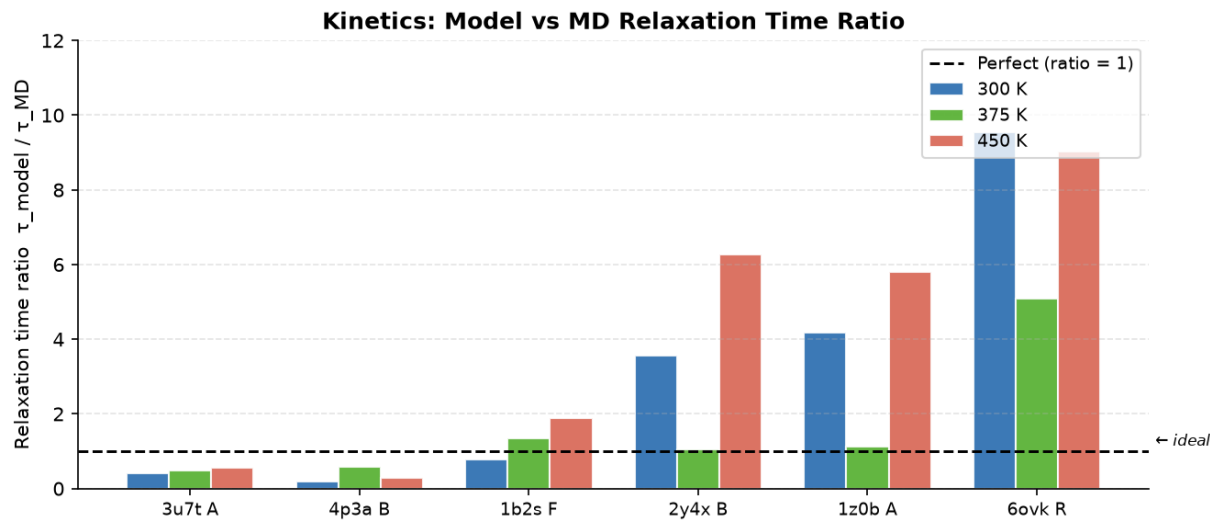


Figure 5. Kinetic relaxation time ratio (model / MD) at each inference temperature. The dashed line marks ideal ratio = 1. Values below 1 indicate over-fast relaxation (model explores conformational space too quickly); values above 1 indicate under-fast.

5.5 Structural Geometry

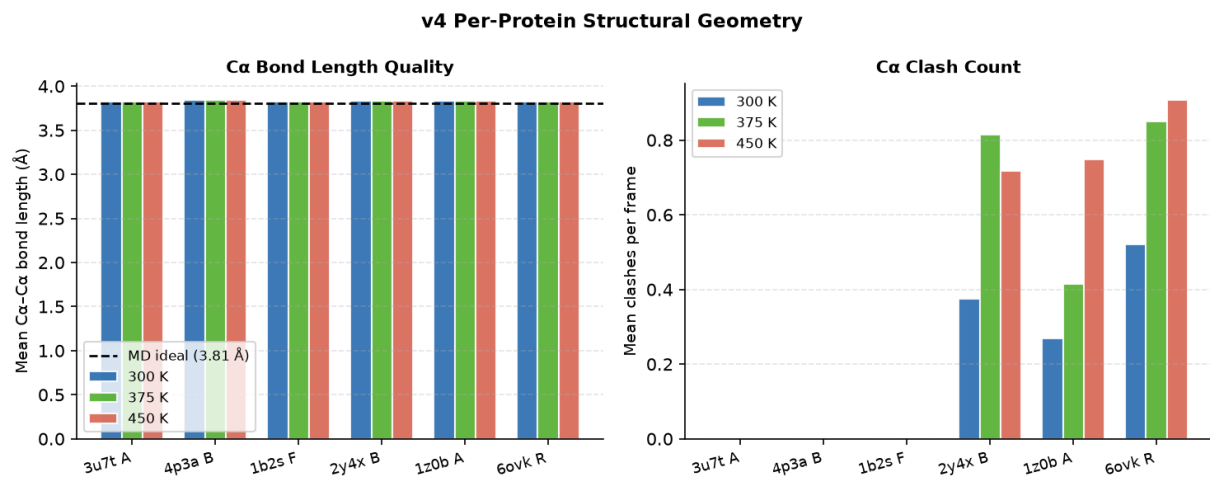


Figure 6. Cα bond length (left) and clash count (right) across proteins and temperatures. The dashed line marks the ideal MD bond length of 3.81 Å. Near-zero clashes confirm the WCA guidance is effective at all temperatures.

5.6 Complete Results Table

Protein	Residues	T _{best} (K)	RMSF corr ↑	Dist JS ↓	FES JS ↓	Relax ratio
3u7t_A	46	375	0.9459	0.000142	0.3330	0.480
4p3a_B	79	375	0.9667	0.000945	0.3815	0.576
1b2s_F	90	300	0.9686	0.000032	0.3756	0.791

2y4x_B	93	375	0.9581	0.000606	0.5481	1.048
1z0b_A	207	300	0.9828	0.000022	0.5605	4.169
6ovk_R	219	375	0.7151	0.000954	0.4878	5.092

Key observations:

- **Smaller proteins** (3u7t_A, 4p3a_B, 1b2s_F) achieve the highest RMSF correlations (0.94–0.97), consistent with their simpler energy landscapes and faster equilibration.
- **Medium proteins** (2y4x_B, 1z0b_A) also perform well (0.96, 0.98), suggesting the per-protein fine-tune effectively captures individual conformational signatures.
- **Large receptor 6ovk_R** (219 residues) achieves $r = 0.72$ — substantially weaker, likely due to slower equilibration timescales that exceed the 5000-step per-protein fine-tune budget and the 300-step validation rollout length.
- **Optimal temperature** is 300–375 K for most proteins. 450 K universally degrades structural metrics, suggesting the temperature embedding amplifies fluctuations past the physically reasonable regime.
- **Kinetics** are consistently under-estimated ($\text{relax_ratio} < 1$ for small proteins) — the model relaxes $\sim 2\text{--}5\times$ faster than MD. This is a known limitation of short per-protein fine-tunes: the model has not learned the full free-energy barriers.

6. V4 Pipeline Summary

Phase	Script / command	Output	Wall time
Phase 1 train	<code>train_transfer.py</code> , 20k steps, all 6 ATLAS shards	<code>v4_longlags.pt</code>	~149 min
Phase 1 validate	<code>validate_physics.py</code> , 300 steps, $T=300$ K	<code>validation_v4_longlags_T300.json</code>	~5 min
Phase 2 train ($\times 6$)	<code>train_transfer.py</code> , 5k steps, single shard	<code>v4_{protein}.pt</code>	~48 min/ protein
Phase 2 validate ($\times 6 \times 3$)	<code>validate_physics.py</code> , $T=\{300,375,450\}$ K	18 JSON files	~4 min/ run
Total	7 checkpoints, 19 JSON files		~5 h 20 min

7. Conclusions and Next Steps

The v4 pipeline demonstrates that combining a wide-lag universal fine-tune with per-protein adaptation and inference temperature tuning yields dramatically better structural and thermodynamic agreement with MD than the v3 baseline. Five of six proteins now achieve RMSF correlation > 0.93 .

Remaining limitations and suggested next steps:

- **Kinetic accuracy:** relaxation ratios are systematically < 1 (too fast). Increasing the per-protein fine-tune budget (5k $\rightarrow$ 20k steps) or adding a kinetic loss term may correct this.
- **6ovk_R:** the large receptor domain underperforms. A longer fine-tune (10k+ steps) and/or a larger per-shard batch may help; the 219-residue chain also has slower intrinsic dynamics that a 300-step rollout may not fully capture.
- **Held-out evaluation:** all ATLAS proteins were seen during training. A proper generalization assessment requires evaluation on proteins not present in the fine-tuning shard set.
- **Multi-chain systems:** the current CA-only model does not capture inter-chain contacts; extending to multi-chain complexes is a logical next step.
- **Explicit free-energy barriers:** training on longer lags (up to 50k ps) begins to capture barrier crossing, but the learned barriers may be imprecise. Coupling with an explicit CG energy model (Phase 3 planned) could improve kinetic fidelity.